iLearn Practice Exam

10BR · Chapter / Topic 1

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 1: Data Representation

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 73

Marks obtained: ..... / 73

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 73.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 Data can be measured in bits.

(a) Give the name of the data storage measurement that is equal to 8 bits.

..... [1]

(b) State how many bits there are in a kibibyte (KiB).

..... [1]

(c) Give the name of the data storage measurement that is equal to 1024 gibibytes (GiB).

..... [1]

(d) A 16-bit colour image has a resolution of 512 pixels wide by 512 pixels high.

Calculate the file size of the image in kibibytes (KiB). Show all your working.

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.....

Answer ..... KiB

[3]

2 Binary is a base 2 number system.

(a) Give the name of the number system that is base 16.

..... [1]

(b) **Three** denary numbers are entered into a computer. The computer converts the numbers and stores them as binary.

(i) Give the binary number that would be stored for each of the denary numbers.

10 .....

50 .....

201 .....

[3]

Working space

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(ii) Explain why the data is converted to binary by the computer.

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..... [2]

(c) The two binary integers 00110000 and 01100110 are added together.

Add the binary integers using binary addition and show your answer in binary. Show all your working.

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..... [3]

(d) The denary integer –32 is stored as a two's complement integer.

Calculate the two's complement integer that would be stored.

Show all your working.

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..... [2]

3 A computer stores data temporarily in random access memory (RAM).

(a) The denary numbers 19 and 230 are converted to binary numbers to be stored in the RAM.

Convert the **two** denary numbers to binary numbers.

19 .....

230 .....

[2]

Working space

.....

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(b) The hexadecimal numbers 35 and 8AD are converted to binary numbers to be stored in the RAM.

Convert the **two** hexadecimal numbers to binary numbers.

35 .....

8AD .....
[2]

Working space

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(c) An overflow occurs when two other 8-bit binary numbers are added together.

Explain why the overflow error occurs.

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..... [2]

(d) The negative denary number –22 is stored in RAM.

Negative denary numbers can be represented as binary using two's complement.

Give the two's complement 8-bit binary integer that would be stored for the denary number –22.

You must show all your working.

Working space .....

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Two's complement 8-bit binary integer .....
[2]

4 A student has an image file stored on their computer.

(a) The size of the image file is 3072 bytes.

(i) Give the size of the image file in kibibytes (KiB).

..... [1]

(ii) State the number of nibbles in 1 byte.

..... [1]

Working space

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(b) The image file has a colour depth of 32 bits.

(i) State what is meant by a colour depth of 32 bits.

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..... [1]

(ii) The colour depth of the image file is changed to 64 bits.

Explain what effect this change has on the image file.

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(c) The student uses lossless compression to compress the image file.

Explain how the image file is compressed using lossless compression.

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..... [3]

5 Data storage can be measured in different units.

(a) Tick (✓) **one** box to show which is the largest unit of measurement for data storage.

A exbibyte (EiB) ☐

B gibibyte (GiB) ☐

C mebibyte (MiB) ☐

D tebibyte (TiB) ☐

[1]

(b) Calculate how many kibibytes (KiB) are equal to 3072 bytes.

..... [1]

working space

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(c) Data is converted to binary to be stored by a computer.

Tick (✓) **one** box to show the correct statement about why data is converted to binary for storing in a computer.

A Binary uses numbers and characters to store all the data. ☐

B Binary is one type of data that a computer can store. ☐

C Data is processed by logic gates that can only be set as binary numbers. ☐

D Only binary numbers can be input into a computer. ☐

[1]

6 Hexadecimal and binary are two examples of a number system.

(a) Give **one** other example of a number system.

..... [1]

(b) Convert the hexadecimal number 2A to an 8-bit binary number.

..... [1]

working space

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(c) Convert the hexadecimal number 101 to a 12-bit binary number.

..... [1]

working space

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(d) Convert the binary number 01110001 to a hexadecimal number.

..... [1]

working space

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(e) Convert the binary number 001011011000 to a hexadecimal number.

..... [1]

working space

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(f) Add the binary numbers 10011101 and 11110000 using binary addition.

Give your answer in binary. Show all your working.

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  1 0 0 1 1 1 0 1
+ 1 1 1 1 0 0 0 0
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[4]

7 A student opens a document on their computer and inputs text.

(a) Explain how the computer represents the text.

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..... [3]

(b) The student stores a video on their computer.

The video contains sound and images.

(i) Complete the table with the missing terms and descriptions about sound and images.

term	description
.....	the smallest unit of an image that is made up of one colour
sound sample	....
.....	the dimensions of an image
colour depth	....
.....	the number of samples that are recorded each second

[5]

(ii) The student wants to compress the video.

Give **two** methods of compression that the student could use to compress the video.

1 .....

2 .....

[2]

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8 A smartwatch has a built-in microphone that allows a person to use voice commands to control the smartwatch.

(a) The smartwatch stores data that is displayed for the user.

Identify the smallest unit of data that is stored by the smartwatch.

..... [1]

(b) One item of data displayed on the smartwatch is the current outdoor temperature. The temperature is stored in a register as a two's complement binary integer.

(i) Tick (✓) **one** box to show whether two's complement is a base 2, base 8, base 10 or base 16 number system.

A base 2 ☐

B base 8 ☐

C base 10 ☐

D base 16 ☐

[1]

(ii) Give the two's complement 8-bit binary integer that would be stored for the temperature of 32 degrees Celsius.

..... [1]

(iii) Give the two's complement 8-bit binary integer that would be stored for the temperature of –14 degrees Celsius.

..... [1]

working space

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9 A video streaming service uses hexadecimal to display error codes when problems occur with their videos.

(a) Tick (✓) **one** box to show which statement is incorrect about why hexadecimal is used to display the error codes.

A Hexadecimal is easier to debug than binary. ☐

B Hexadecimal is easier to read than binary. ☐

C Hexadecimal uses less screen space than binary. ☐

D Hexadecimal uses less storage space than binary. ☐

[1]

(b) There is a problem with streaming a video and the error codes 152 and 30F are displayed.

Convert the **two** hexadecimal error codes to binary numbers.

152 .....

30F .....

[2]

working space

.....

.....

.....

(c) The binary numbers 010001110101 and 101011101000 are stored.

Convert the **two** binary numbers to the hexadecimal error codes that would be displayed.

010001110101 .....

101011101000 .....

[2]

working space

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(d) Hexadecimal numbers can also be converted to denary numbers.

Convert the **two** hexadecimal error codes 0A2 and 104 to denary numbers.

0A2 .....

104 .....

[2]

working space

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(e) The video streaming service compresses their videos using lossless compression.

(i) Give **two** reasons why the video streaming service chose lossless compression instead of lossy compression.

1 .....

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2 .....

..... [2]

(ii) Give **one** advantage to the video streaming service of choosing lossy compression instead of lossless compression to compress the videos.

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..... [1]

(iii) Explain how the videos are compressed using lossless compression.

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..... [4]

END OF QUESTION PAPER